

# Boyu Liu

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## EDUCATION

Carnegie Mellon University

M.S. in Information Networking

Aug 2026 – May 2028

University of Illinois Urbana-Champaign

BS in Mathematics and Computer Science

Aug 2022 – May 2026

- **GPA: 3.88/4.0** | Graduated with Highest Distinction
- **Coursework:** Data Structures, Linear Algebra, Computer Systems, Algorithms, Numerical Methods, Machine Learning, Database Systems, Machine Learning Systems, Distributed Systems, AI Agent, Compilers

## WORK EXPERIENCE

TetraBot

R&D Intern

Nanjing, China

June 2026 – Aug 2026

- Re-architected a legacy LLM-agent-orchestrated fire-response system into a deterministic state-machine kernel with deploy-time AI agents using **Python**, cutting incident handling from **497.5s to 167.9s (-66%)**
- Built the runtime on **Python asyncio**, **Redis Streams**, **SQLite**, decreasing first-response latency from **19s to 36ms** with **3.7× throughput** under concurrent alarms, zero-loss **kill -9** crash recovery, and SLA-timeout red lines
- Implemented a lightweight **agent runtime** with bounded tool-calling loops, **JSON-Schema** validation, per-persona permission whitelists, and resumable on-disk sessions and audit logging
- Designed a workflow **DSL** and an **LLM compiler** that writes each building's emergency-response plan, then auto-verifies it against safety rules and simulated fire drills in a **generate-verify-repair** loop
- Built the user-facing layer end-to-end: a conversational **ingestion agent** turning messy device sheets, floor-plan images, and CAD drawings into human-confirmed building facts, cutting onboarding from days of scripting to one dialogue session; also shipped a **voice copilot** and a **React + Three.js** 3D building view with live fire localization
- Deployed **YOLOv8n** fire/smoke detection on the RK3588 NPU via **ONNX** → **RKNN INT8 quantization**, which increased FPS from **2** → **41 (20×)** at FP16-equivalent accuracy, served over HTTP into the multi-frame analysis chain

Machine Intelligence Lab @ Westlake University

Research Intern

Hangzhou, China

Jul 2025 – Sep 2025

- Researched **offline RL** for long-horizon, high-precision robotic manipulation on FurnitureBench furniture-assembly tasks
- Built the end-to-end training and evaluation pipeline with **Python**, **PyTorch**, and **IsaacGym** simulation, adapting the **Reinformer** sequence-modeling architecture and fusing multi-modal inputs for high-dimensional control
- Implemented **Max-Return Sequence Modeling**, predicting in-distribution maximum returns via **expectile regression** to address the **trajectory stitching** failure mode of standard Decision Transformers, and stabilized optimization with entropy-temperature tuning and gradient clipping

National Center for Supercomputing Applications

Full-Stack Software Engineer

Champaign, IL

Oct 2024 – Dec 2024

- Built EACUE, a mobile-first full-stack survey platform for recording accessibility evaluations across physical venues, replacing fragmented manual collection workflows with a centralized digital system
- Engineered a **Next.js**, **Node.js**, and **MongoDB** architecture to support end-to-end survey delivery, structured data storage, and **CRUD** workflows for users, venues, and evaluation records
- Designed and implemented a **dynamic question-routing** engine that adapted survey flow based on **real-time user selections**, improving form efficiency and reducing unnecessary input steps

## PROJECTS

IRTS-ToolBench + TSIR-Agent — Benchmark and Verifiable Agent for Irregular Time-Series QA

- Built IRTS-ToolBench (arxiv.org/abs/2606.15107), the largest benchmark for irregular time-series QA agents, with 20K questions across 10 task types and 13 domains to evaluate reasoning under asynchronous sampling, missingness, and structural irregularity
- Developed a specialized irregularity-aware tool library for alignment, missingness handling, regularity recovery, and downstream time-series analysis
- Designed TSIR-Agent, a process-verifiable multi-agent framework with Observation, Planning, Execution, and Verification stages plus schema-constrained tool use for auditable reasoning

- Outperformed baseline LLMs by **40 percentage points** on the benchmark, establishing a new leading result on our proposed irregular TSQA benchmark

### Rhythm — Voice-First iOS Productivity App

- Built a voice-first task management iOS app using **SwiftUI** and **MVVM** architecture, enabling users to capture tasks through natural-language speech input and manage them through a flexible, low-friction planning workflow
- Architected a modular Python backend for **agent orchestration** using FastAPI, separating API, tool runtime, service, and repository layers to support **structured tool calling** for task creation, context retrieval, and status updates
- Designed and deployed secure backend infrastructure with **Supabase** utilizing **PostgreSQL** with Row Level Security policies, **JWT authentication**, and **RESTful APIs** to ensure user data isolation
- Developed intelligent background sync service with conflict resolution, retry logic, and network state monitoring, enabling 100% offline functionality with automatic reconciliation upon reconnection
- Leveraged **Swift Concurrency**, **AppIntents**, and **Apple Speech APIs** to support responsive voice interactions, background task handling, and Siri-triggered capture on iOS

### Raft-Based Distributed Log Replication System

- Implemented a fault-tolerant **Raft** consensus module in **Go**, supporting **leader election**, **randomized election timeouts**, **heartbeats**, and **term-based state transitions** across replicated server instances
- Built log replication and commitment logic using **RequestVote** and **AppendEntries** RPCs, maintaining per-follower replication state, majority-based commit advancement, and ordered **ApplyMsg** delivery to the service layer
- Validated correctness under simulated distributed failures, including leader crashes, follower disconnections, network partitions, delayed/reordered/dropped RPCs, concurrent client commands, and rejoining nodes with inconsistent logs

### E<sup>3</sup> Mini-Benchmark

- Designed and implemented a comprehensive **benchmarking framework** for Small Language Models (SLMs) centered on the Efficiency-Energy-Effectiveness triad
- Implemented **LoRA** parameter-efficient continuous pretraining and fine-tuning with architecture-specific target modules, reducing trainable parameters to **2.4%** and GPU memory by **80%**
- Engineered a hardware-aware Inference Engine measuring **TTFT** & **TBT** Latency, Throughput, and VRAM; integrated a custom nvidia-smi based PowerMonitor to calculate precise Energy-Per-Token metrics
- Built an automated evaluation suite combining **SuperGLUE** fine-tuning (handling both classification and Seq2Seq heads) and **MMLU** Few-Shot tasks to assess Natural Language Understanding and in-context learning capabilities
- Conducted a systematic comparison of Encoder-only, Decoder-only, and Encoder-Decoder architectures, quantitatively benchmarking their performance, speed, and energy consumption trade-offs

## TECHNICAL SKILLS

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- **Programming Languages:** Python, C++, C, Java, JavaScript, TypeScript, Swift, Golang, SQL, CUDA
- **ML & Data:** PyTorch, TensorFlow, TensorBoard, NumPy, Pandas, Matplotlib, LangChain, W&B, Verl, SGLang
- **Web:** React.js, Svelte, HTML/CSS, Spring Boot, FastAPI, Express.js, Node.js, Next.js, MongoDB, MySQL, PostgreSQL
- **Cloud and Others:** AWS, GCP, Postman, Git, GitHub, Docker, Vercel, Supabase, Notion

## PUBLICATIONS

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Baoyu Jing, Sanhorn Chen, Lecheng Zheng, **Boyu Liu**, Zihao Li, et al. "TSAQA: Time Series Analysis Question and Answering Benchmark." *The 5th Workshop on Generation, Evaluation Metrics (GEM 2026)*, *ACL 2026*. arxiv.org/abs/2601.23204

**Boyu Liu**. "Comparative Analysis of Encoder-Only, Decoder-Only, and Encoder-Decoder Language Models." doi:10.5220/0012829800004547